

In [8]:

```
!pip install pandas
```

Requirement already satisfied: pandas in c:\users\rgukt\anaconda3\anak2\lib\site-packages (0.23.4)
Requirement already satisfied: python-dateutil>=2.5.0 in c:\users\rgukt\anaconda3\anak2\lib\site-packages (from pandas) (2.7.3)
Requirement already satisfied: pytz>=2011k in c:\users\rgukt\anaconda3\anak2\lib\site-packages (from pandas) (2018.5)
Requirement already satisfied: numpy>=1.9.0 in c:\users\rgukt\anaconda3\anak2\lib\site-packages (from pandas) (1.15.1)
Requirement already satisfied: six>=1.5 in c:\users\rgukt\anaconda3\anak2\lib\site-packages (from python-dateutil>=2.5.0->pandas) (1.11.0)

twisted 18.7.0 requires PyHamcrest>=1.9.0, which is not installed.
You are using pip version 10.0.1, however version 22.2.2 is available.
You should consider upgrading via the 'python -m pip install --upgrade pip' command.

In []:

```
!conda install pandas
```

In []:

```
!conda install pandas
```

In []:

```
import pandas ps
```

In []:

```
df=ps.read_csv("ref.csv")
```

In []:

```
df
```

In []:

```
from pandas import read_csv
```

In [2]:

```
import pandas as ps  
df=ps.read_csv("students.csv")
```

In [6]:

```
df.head(n=10)
```

Out[6]:

	ID	CN	ES	ML	LIC
0	N160380	27	29	33	34
1	N160918	25	22	33	33
2	N160999	28	33	32	33
3	N170016	29	32	33	33
4	N170017	29	30	32	33
5	N170030	28	22	33	32
6	N170043	30	32	32	33
7	N170057	33	34	32	33
8	N170082	33	29	32	32
9	N170088	33	26	33	32

In [44]:

```
df1=df.iloc[0:5,0:3]
```

In [45]:

```
df1
```

Out[45]:

	ID	CN	ES
0	N160380	27	29
1	N160918	25	22
2	N160999	28	33
3	N170016	29	32
4	N170017	29	30

In [46]:

```
df.loc[0:7,["CN","ES"]]
```

Out[46]:

	CN	ES
0	27	29
1	25	22
2	28	33
3	29	32
4	29	30
5	28	22
6	30	32
7	33	34

In [11]:

```
#df["CN"]>25  
df1=df.loc[df["CN"]>25,["ES","CN"]]
```

In [12]:

```
df1
```

Out[12]:

	ES	CN
0	29	27
2	33	28
3	32	29
4	30	29
5	22	28
6	32	30
7	34	33
8	29	33
9	26	33
10	34	30
11	34	30
12	25	29
13	22	32
14	33	26
16	28	30
17	28	30
18	30	29
19	26	30
20	32	29
23	31	33
24	28	29
26	33	30
28	24	31
29	30	34
30	30	32
31	26	28
32	29	30
33	30	34
34	33	27
35	25	29
36	26	30
38	26	31
39	32	26
40	30	31
41	30	26
42	31	31
43	29	30
44	28	29
46	32	30
47	24	35
48	30	35

In [52]:

```
df2=df[df["ES"]>30]
```

In [53]:

```
df2
```

Out[53]:

	ID	CN	ES	ML	LIC
2	N160999	28	33	32	33
3	N170016	29	32	33	33
6	N170043	30	32	32	33
7	N170057	33	34	32	33
10	N170103	30	34	32	32
11	N170109	30	34	35	33
14	N170143	26	33	32	32
15	N170160	25	33	32	34
20	N170201	29	32	33	25
22	N170206	22	32	33	35
23	N170241	33	31	32	34
25	N170283	25	32	32	33
26	N170302	30	33	32	33
34	N170476	27	33	33	34
39	N170677	26	32	33	34
42	N170760	31	31	33	33
46	N170898	30	32	32	28

In [55]:

```
df[df["ES"].isin([31,32])]
```

Out[55]:

	ID	CN	ES	ML	LIC
3	N170016	29	32	33	33
6	N170043	30	32	32	33
20	N170201	29	32	33	25
22	N170206	22	32	33	35
23	N170241	33	31	32	34
25	N170283	25	32	32	33
39	N170677	26	32	33	34
42	N170760	31	31	33	33
46	N170898	30	32	32	28

In [62]:

```
df3=df[df["ES"].isin([31,32])]
```

In [63]:

```
df3
```

Out[63]:

	ID	CN	ES	ML	LIC
3	N170016	29	32	33	33
6	N170043	30	32	32	33
20	N170201	29	32	33	25
22	N170206	22	32	33	35
23	N170241	33	31	32	34
25	N170283	25	32	32	33
39	N170677	26	32	33	34
42	N170760	31	31	33	33
46	N170898	30	32	32	28

In [64]:

```
df3.reset_index(inplace=True)
```

In [65]:

```
df3
```

Out[65]:

	index	ID	CN	ES	ML	LIC
0	3	N170016	29	32	33	33
1	6	N170043	30	32	32	33
2	20	N170201	29	32	33	25
3	22	N170206	22	32	33	35
4	23	N170241	33	31	32	34
5	25	N170283	25	32	32	33
6	39	N170677	26	32	33	34
7	42	N170760	31	31	33	33
8	46	N170898	30	32	32	28

In [14]:

```
df.describe()
```

Out[14]:

	CN	ES	ML	LIC
count	49.000000	49.000000	49.000000	49.000000
mean	29.061224	29.020408	32.734694	31.469388
std	3.337964	3.350931	0.836050	3.055746
min	20.000000	22.000000	32.000000	22.000000
25%	27.000000	26.000000	32.000000	31.000000
50%	30.000000	30.000000	33.000000	33.000000
75%	31.000000	32.000000	33.000000	33.000000
max	35.000000	34.000000	35.000000	35.000000

In [67]:

```
import numpy as np
```

In [69]:

```
df["grade"]=np.where(df["ES"]>30,"grade A","grade B")
```

In [70]:

```
df.head()
```

Out[70]:

	ID	CN	ES	ML	LIC	grade
0	N160380	27	29	33	34	grade B
1	N160918	25	22	33	33	grade B
2	N160999	28	33	32	33	grade A
3	N170016	29	32	33	33	grade A
4	N170017	29	30	32	33	grade B

In [71]:

```
df
```

Out[71]:

	ID	CN	ES	ML	LIC	grade
0	N160380	27	29	33	34	grade B
1	N160918	25	22	33	33	grade B
2	N160999	28	33	32	33	grade A
3	N170016	29	32	33	33	grade A
4	N170017	29	30	32	33	grade B
5	N170030	28	22	33	32	grade B
6	N170043	30	32	32	33	grade A
7	N170057	33	34	32	33	grade A
8	N170082	33	29	32	32	grade B
9	N170088	33	26	33	32	grade B
10	N170103	30	34	32	32	grade A
11	N170109	30	34	35	33	grade A
12	N170120	29	25	32	30	grade B
13	N170125	32	22	32	22	grade B
14	N170143	26	33	32	32	grade A
15	N170160	25	33	32	34	grade A
16	N170167	30	28	32	29	grade B
17	N170192	30	28	33	26	grade B
18	N170193	29	30	35	34	grade B
19	N170198	30	26	34	34	grade B
20	N170201	29	32	33	25	grade A
21	N170203	23	28	33	33	grade B
22	N170206	22	32	33	35	grade A
23	N170241	33	31	32	34	grade A
24	N170273	29	28	33	33	grade B
25	N170283	25	32	32	33	grade A
26	N170302	30	33	32	33	grade A
27	N170310	23	28	33	32	grade B
28	N170354	31	24	33	33	grade B
29	N170386	34	30	32	29	grade B
30	N170414	32	30	32	25	grade B
31	N170438	28	26	32	30	grade B
32	N170459	30	29	32	23	grade B
33	N170460	34	30	33	31	grade B
34	N170476	27	33	33	34	grade A
35	N170477	29	25	33	33	grade B
36	N170509	30	26	32	34	grade B
37	N170634	25	25	33	33	grade B
38	N170638	31	26	34	33	grade B
39	N170677	26	32	33	34	grade A
40	N170715	31	30	33	33	grade B
41	N170747	26	30	34	33	grade B
42	N170760	31	31	33	33	grade A
43	N170809	30	29	33	32	grade B
44	N170828	29	28	33	32	grade B
45	N170892	20	26	32	29	grade B
46	N170898	30	32	32	28	grade A
47	N170910	35	24	35	26	grade B
48	N170911	35	30	32	32	grade B

In [72]:

```
df.groupby("ID")["ES"].agg("mean")
```

Out[72]:

```
ID
N160380    29
N160918    22
N160999    33
N170016    32
N170017    30
N170030    22
N170043    32
N170057    34
N170082    29
N170088    26
N170103    34
N170109    34
N170120    25
N170125    22
N170143    33
N170160    33
N170167    28
N170192    28
N170193    30
N170198    26
N170201    32
N170203    28
N170206    32
N170241    31
N170273    28
N170283    32
N170302    33
N170310    28
N170354    24
N170386    30
N170414    30
N170438    26
N170459    29
N170460    30
N170476    33
N170477    25
N170509    26
N170634    25
N170638    26
N170677    32
N170715    30
N170747    30
N170760    31
N170809    29
N170828    28
N170892    26
N170898    32
N170910    24
N170911    30
Name: ES, dtype: int64
```

In [73]:

```
g1=df.groupby("ID")["ES"].agg("mean")
```

In [74]:

```
g1.to_csv("stud.csv")
```

In [75]:

```
df.head()
```

Out[75]:

	ID	CN	ES	ML	LIC	grade
0	N160380	27	29	33	34	grade B
1	N160918	25	22	33	33	grade B
2	N160999	28	33	32	33	grade A
3	N170016	29	32	33	33	grade A
4	N170017	29	30	32	33	grade B

In [84]:

```
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

In []:

In [81]:

```
df.head()
```

Out[81]:

	ID	CN	ES	ML	LIC	grade
0	N160380	27	29	33	34	grade B
1	N160918	25	22	33	33	grade B
2	N160999	28	33	32	33	grade A
3	N170016	29	32	33	33	grade A
4	N170017	29	30	32	33	grade B

In [82]:

```
df.describe(include="all")
```

Out[82]:

	ID	CN	ES	ML	LIC	grade
count	49	49.000000	49.000000	49.000000	49.000000	49
unique	49	NaN	NaN	NaN	NaN	2
top	N170203	NaN	NaN	NaN	NaN	grade B
freq	1	NaN	NaN	NaN	NaN	32
mean	NaN	29.061224	29.020408	32.734694	31.469388	NaN
std	NaN	3.337964	3.350931	0.836050	3.055746	NaN
min	NaN	20.000000	22.000000	32.000000	22.000000	NaN
25%	NaN	27.000000	26.000000	32.000000	31.000000	NaN
50%	NaN	30.000000	30.000000	33.000000	33.000000	NaN
75%	NaN	31.000000	32.000000	33.000000	33.000000	NaN
max	NaN	35.000000	34.000000	35.000000	35.000000	NaN

In [83]:

```
df.info()
```

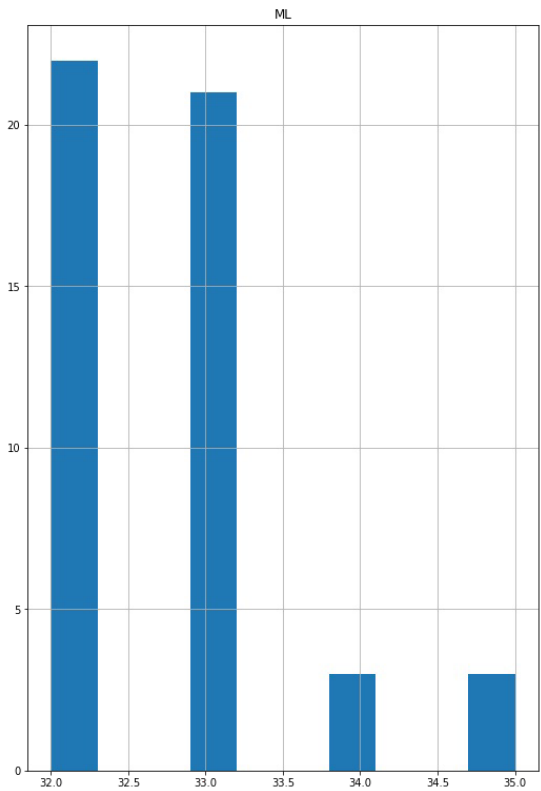
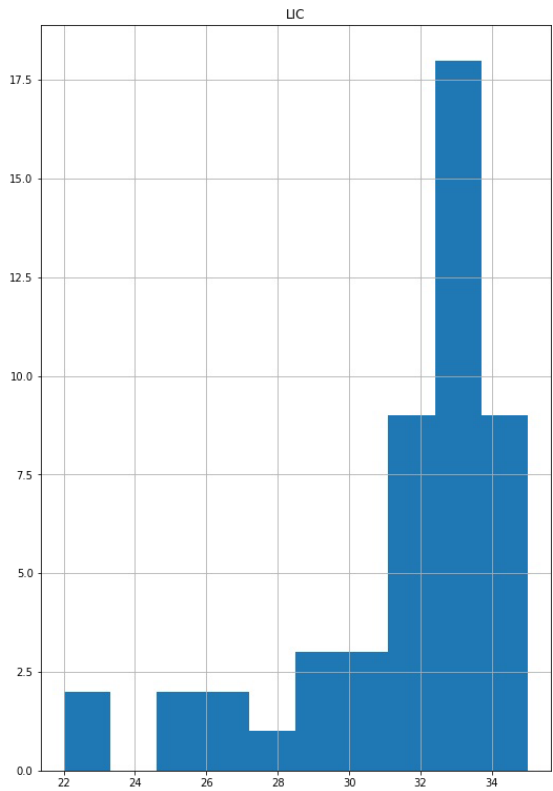
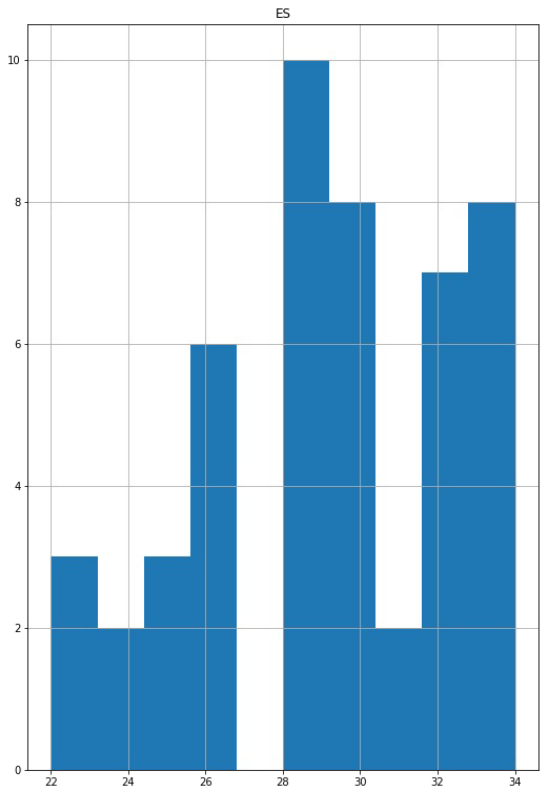
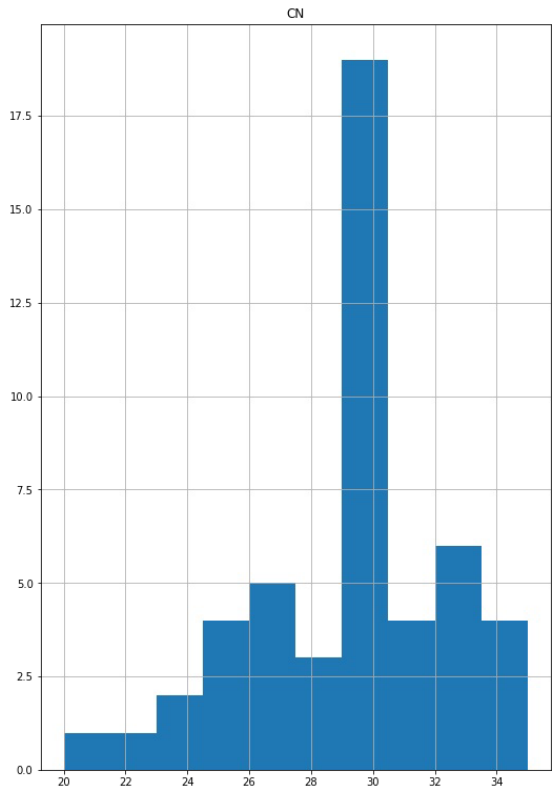
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49 entries, 0 to 48
Data columns (total 6 columns):
ID      49 non-null object
CN      49 non-null int64
ES      49 non-null int64
ML      49 non-null int64
LIC     49 non-null int64
grade   49 non-null object
dtypes: int64(4), object(2)
memory usage: 2.0+ KB
```

In [87]:

```
import matplotlib.pyplot as plt
%matplotlib inline
df.hist(figsize=(20,30))
```

Out[87]:

```
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x080CA950>,  
      <matplotlib.axes._subplots.AxesSubplot object at 0x086E92B0>],  
      [<matplotlib.axes._subplots.AxesSubplot object at 0x08947210>,  
      <matplotlib.axes._subplots.AxesSubplot object at 0x08964170>]],  
      dtype=object)
```

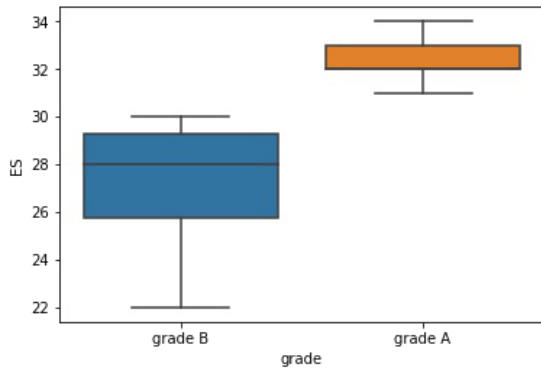


In [90]:

```
import seaborn as sb
sb.boxplot(x="grade",y="ES",data=df)
```

Out[90]:

<matplotlib.axes._subplots.AxesSubplot at 0x8901d90>

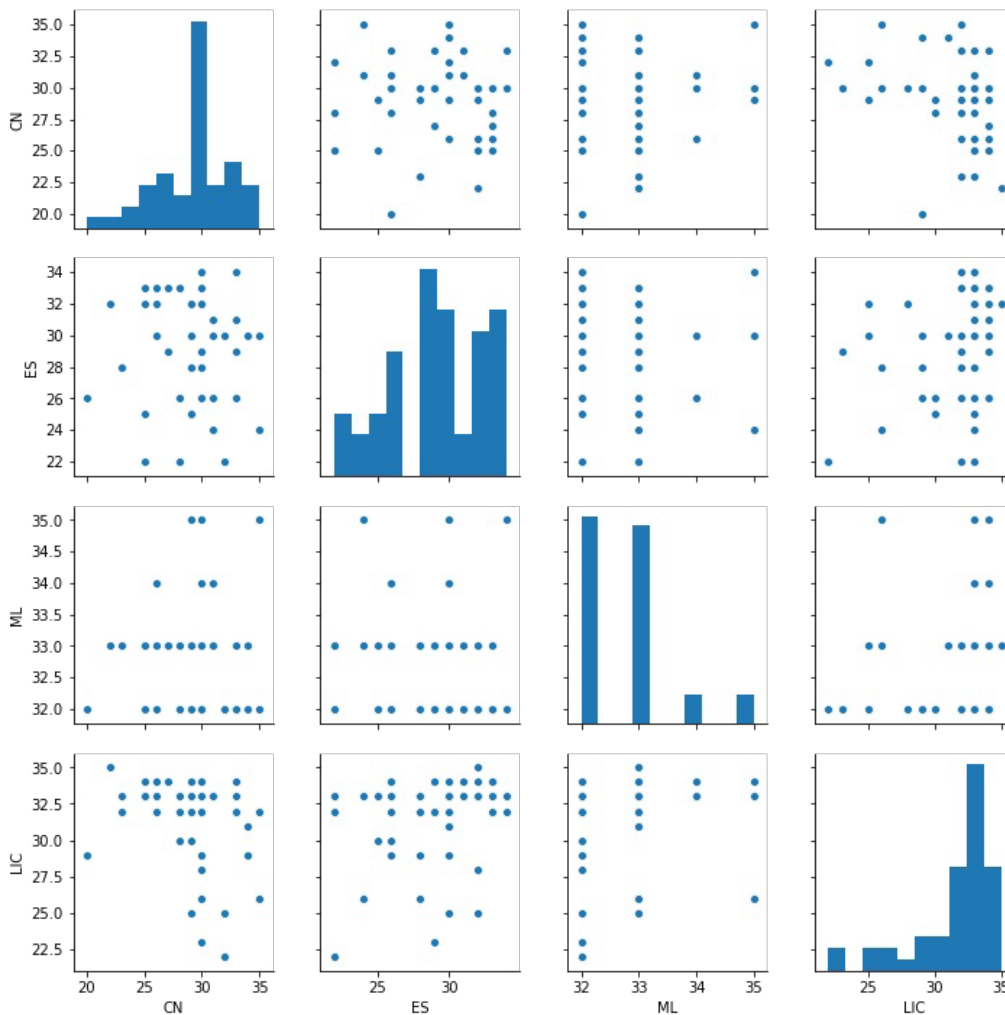


In [91]:

```
#univariant (mirror images)
sns.pairplot(df)
```

Out[91]:

<seaborn.axisgrid.PairGrid at 0x8c238b0>



In [92]:

```
df.drop(['ID'],axis='columns',inplace=True)
```

In [93]:

```
df.head()
```

Out[93]:

	CN	ES	ML	LIC	grade
0	27	29	33	34	grade B
1	25	22	33	33	grade B
2	28	33	32	33	grade A
3	29	32	33	33	grade A
4	29	30	32	33	grade B

In [95]:

```
out1=df.ES  
Inputs=df.drop('ES',axis='columns')
```

In [98]:

```
dummies=pd.get_dummies(Inputs.grade)  
dummies.head()
```

Out[98]:

	grade A	grade B
0	0	1
1	0	1
2	1	0
3	1	0
4	0	1

In [100]:

```
Inputs=pd.concat([Inputs,dummies],axis='columns')  
Inputs.head()
```

Out[100]:

	CN	ML	LIC	grade	grade A	grade B	grade A	grade B
0	27	33	34	grade B	0	1	0	1
1	25	33	33	grade B	0	1	0	1
2	28	32	33	grade A	1	0	1	0
3	29	33	33	grade A	1	0	1	0
4	29	32	33	grade B	0	1	0	1

In []: